CHAPTER EXERCISE: Answers for this chapter start on page 304.

A calculator is allowed on the following questions.

1

Katherine has 28 classroom calculators that each require a set of 4 batteries. If her school supplies her with batteries in packs of 6, what is the least number of packs needed to provide every classroom calculator with a complete set of batteries?

2

Martha is working on a design project that requires 16 ounces of glue. The glue gun she is using comes preloaded with a glue stick that provides 2.5 ounces of glue. The only additional glue sticks Martha can purchase are ones that each provide 1.75 ounces of glue. Assuming that the glue sticks can only be purchased in whole numbers, what is the minimum number of glue sticks Martha must purchase for her project?

A) 6
B) 7
C) 8
D) 9

3

A gift shop held a weekend sale with the goal of selling at least $8,000 worth of greeting cards and gift boxes. Each greeting card was sold for $5, and each gift box was sold for $7. If no more than 400 gift boxes were sold during the sale due to limited inventory, what is the minimum number of greeting cards the shop could have sold to meet its goal?

A) 1,040
B) 1,160
C) 1,280
D) 1,400

4

To restock supplies, a nail salon purchases toolkits that each include 80 nail files and 150 nail buffers. If the nail salon needs to restock at least 1,800 nail files and at least 4,000 nail buffers, what is the minimum number of toolkits the salon can purchase?

5

One liter is equivalent to approximately 33.8 ounces. Mark has plastic cups that can each hold 12 ounces of liquid. At most, how many of these plastic cups could a two liter bottle of soda fill?

A) 5
B) 6
C) 7
D) 8

6

In one hour, Jason can install at least 6 windows but no more than 8 windows. Which of the following could be a possible amount of time, in hours, that Jason takes to install 100 windows in a home?

A) 12
B) 16
C) 17
D) 18

7

1 fluid ounce = 29.6 milliliters
1 cup = 16 fluid ounces

A chemistry teacher is planning to run a class experiment in which each student must measure out 100 milliliters of vinegar in a graduated cylinder. The class is limited to using 6 cups of vinegar. Given the information above, what is the maximum number of students who will be able to participate in this experiment?

8

Giovanni works as a waiter at an Italian restaurant. For every table that he serves, he earns a 15% tip on the bill. During lunch, he served 12 tables and each table had an average bill of $25. If each table during dinner will have an average bill of $45, what is the least number of tables Giovanni must serve during dinner to earn at least $180 for the day?

A) 3
B) 16
C) 18
D) 20

9

During a week-long fishing trip, Ashleigh caught nine less than three times the number of fish Naomi caught. If they caught at least 45 fish combined, what is the minimum number of fish that Naomi could have caught?

10

A jar is filled with black pebbles, white pebbles, and jade pebbles. The number of jade pebbles is greater than half the number of black pebbles, and the number of white pebbles is less than twice the number of black pebbles. If there are 32 jade pebbles in the jar, what is the maximum number of white pebbles that could be in the jar?

11

A pharmacy produces a certain medication in a daytime variety and a nighttime variety. A bottle of the daytime variety contains 2 ounces of the active ingredient and 6 ounces of flavored syrup. A bottle of the nighttime variety contains 3 ounces of the active ingredient and 5 ounces of flavored syrup. The pharmacy currently has no more than 385 ounces of the active ingredient and no more than 850 ounces of flavored syrup available. If at least 65 bottles of the daytime variety must be filled, what is the maximum number of bottles of the nighttime variety that can be filled?

A) 78
B) 85
C) 92
D) 106

12

A banquet hall has a maximum seating capacity of 168 people. For a particular event, the banquet manager must use an arrangement of short tables and long tables to ensure that there is enough seating to meet that capacity. Each short table seats 4 people and each long table seats 8 people. If no more than 32 tables can be placed inside the banquet hall, what is the maximum number of short tables that can be used?

A) 10
B) 14
C) 18
D) 22

13

As part of a marketing campaign, a restaurant is offering 4 free tacos for every burrito a customer buys. If the restaurant would normally sell the tacos for $2.60 each, what is the minimum number of burritos a customer would have to buy to receive at least $140 worth of tacos for free?

14

Ava is decorating two-tier and three-tier wedding cakes. It takes her 20 minutes to decorate each two-tier wedding cake and 35 minutes to decorate each three-tier wedding cake. If Ava needs to decorate at least 14 wedding cakes today, and she can spend no more than 6 hours doing so, what is the maximum number of three-tier wedding cakes she can decorate today?

A) 4
B) 5
C) 6
D) 8

15

Lianne wants to make a seasoning that consists of 75% sea salt and 25% black pepper. If sea salt costs \$2 per pound and black pepper costs \$8 per pound, and Lianne can spend no more than \$210 on these ingredients, what is the maximum number of pounds of seasoning that she will be able to make?

A) 42

B) 50

C) 56

D) 60

16

A toy company ships its products in small, medium, and large boxes. Last month, the company shipped a total of 250 boxes, of which 70 were medium boxes. The number of large boxes shipped was more than the sum of the number of small boxes shipped and the number of medium boxes shipped. What is the greatest possible number of small boxes the company shipped last month?

17

$$C = \frac{100n}{n+w}\%$$

The formula above can be used to determine the volume percent concentration C of an ethanol solution containing n ounces of ethanol and w ounces of water. A chemist wants to use the formula to create an ethanol solution with a volume percent concentration of no more than 16%. If the chemist will mix 10 ounces of ethanol and x cups of water to create the desired solution, what is the minimum possible value of x, assuming that x is a whole number?
(1 cup = 8 ounces)

Chapter 13: Minimum & Maximum Word Problems

CHAPTER EXERCISE:

1. $\boxed{19}$ Katherine will need a total of $28 \times 4 = 112$ batteries. Given that there are 6 batteries in a pack, she will need $112 \div 6 \approx 18.67$ packs. Since it's implied that packs only come in whole numbers, she needs to round up to 19 packs to ensure she has enough batteries.

2. $\boxed{C}$ Martha needs $16 - 2.5 = 13.5$ more ounces of glue, which amounts to $\frac{13.5}{1.75} \approx 7.7$ glue sticks. Since glue sticks can only be purchased in whole amounts, Martha must purchase a minimum of 8 glue sticks.

3. $\boxed{A}$ To get the minimum number of greeting cards the shop could have sold, we assume that the shop sold as many gift boxes as possible (400 gift boxes). Since each gift box was sold for \$7, the store sold $400 \times \$7 = \$2,800$ worth of gift boxes in this scenario, which means the store sold at least $\$8,000 - \$2,800 = \$5,200$ worth of greeting cards. Since each greeting card was sold for \$5, the store could have sold a minimum of $\$5,200 \div \$5 = 1,040$ greeting cards to meet its goal.

4. $\boxed{27}$ To get the required number of nail files, the salon needs at least $\frac{1,800}{80} = 22.5 \rightarrow 23$ toolkits. To get the required number of nail buffers, the salon needs at least $\frac{4,000}{150} \approx 26.7 \rightarrow 27$ toolkits. Based on these numbers, a minimum of 27 toolkits must be purchased for the salon to receive both the required number of nail files and the required number of nail buffers.

5. $\boxed{A}$ Two liters is equivalent to $2 \times 33.8 = 67.6$ ounces, which will fill $67.6 \div 12 \approx 5.63$ plastic cups. So at most, 5 plastic cups can be completely filled.

6. $\boxed{B}$ Working at the slowest pace, Jason would take $100 \div 6 \approx 16.67$ hours. Working at the fastest pace, he would take $100 \div 8 = 12.5$ hours. The only answer choice between those two numbers is 16.

7. $\boxed{28}$

$$6 \cancel{\text{cups}} \times \frac{16 \cancel{\text{ounces}}}{1 \cancel{\text{cup}}} \times \frac{29.6 \cancel{\text{mL}}}{1 \cancel{\text{ounce}}} \times \frac{1 \text{ student}}{100 \cancel{\text{mL}}}$$

$$\approx 28.4 \text{ students}$$

Since it wouldn't make sense to have four-tenths of a student, the most that can be accommodated is 28 students.

8. $\boxed{D}$ Giovanni made $0.15(25)(12) = \$45$ during lunch. If he serves x tables during dinner, he will make an additional $0.15(45)x$ dollars. Since the total for the day should be at least \$180,

$$\begin{aligned} 45 + 0.15(45)x &\geq 180 \\ 0.15(45)x &\geq 135 \\ x &\geq \frac{135}{0.15(45)} \\ x &\geq 20 \end{aligned}$$

9. $\boxed{14}$ Let a be the number of fish Ashleigh caught and n be the number of fish Naomi caught. Using these variables, we can set up a system of an equation and an inequality:

$$a = 3n - 9$$
$$a + n \geq 45$$

The equation allows us to substitute for a in the inequality:

$$a + n \geq 45$$
$$(3n - 9) + n \geq 45$$
$$4n \geq 54$$
$$n \geq 13.5$$

Since it's implied that fish are caught in whole numbers, the minimum possible value of n is 14.

10. $\boxed{125}$ If we let b be the number of black pebbles, w be the number of white pebbles, and j be the number of jade pebbles, then $j > \frac{b}{2}$ and $w < 2b$. Since $j = 32$, the first inequality becomes $32 > \frac{b}{2}$, which simplifies to $64 > b$, so the maximum possible value of b is 63. Using $b = 63$, the second inequality becomes $w < 2(63)$, which simplifies to $w < 126$. Based on this result, the maximum possible value of w is 125. Now you might be wondering why we used the maximum possible value of b in the second inequality. Since w is less than $2b$, maximizing w means that we have to maximize b first.

11. $\boxed{B}$ Because we're trying to maximize the number of nighttime bottles, we assume that only 65 daytime bottles were filled. The 65 daytime bottles used up $65 \times 2 = 130$ ounces of the active ingredient and $65 \times 6 = 390$ ounces of flavored syrup, leaving $385 - 130 = 255$ ounces of the active ingredient and $850 - 390 = 460$ ounces of flavored syrup. The remaining ounces of active ingredient are enough for $\frac{255}{3} = 85$ nighttime bottles, and the remaining ounces of flavored syrup are enough for $\frac{460}{5} = 92$ nighttime bottles. Based on these numbers, we're limited by the remaining amount of active ingredient, so the maximum number of nighttime bottles that can be filled is 85.

12. $\boxed{D}$ Let's set up a system with s as the number of short tables and l as the number of long tables.

$$4s + 8l = 168$$
$$s + l \leq 32$$

Divide both sides of the equation by 4 to get $s + 2l = 42$. Isolating l then gives $l = 21 - \frac{s}{2}$. Substituting this result into the inequality, we get

$$s + 21 - \frac{s}{2} \leq 32$$
$$\frac{s}{2} \leq 11$$
$$s \leq 22$$

Based on this result, the maximum number of short tables that can be used is 22.

13. $\boxed{14}$ To get at least \$140 worth of tacos, a customer would have to receive at least $\frac{140}{2.60} \approx 53.8 \rightarrow 54$ tacos (we round up since it's implied that tacos are given in whole numbers only). To receive at least 54 tacos, the customer would have to buy at least $\frac{54}{4} = 13.5 \rightarrow 14$ burritos (again, we round up since it's implied that burritos are sold in whole numbers). Therefore, 14 is the minimum.

14. $\boxed{B}$ Let a be the number of two-tier cakes and b be the number of three-tier cakes Ava decorates. Using these variables, we can set up the following system of inequalities:

$$\begin{aligned} 20a + 35b &\leq 360 \\ a + b &\geq 14 \end{aligned}$$

Note that we converted 6 hours to 360 minutes to set up the first inequality. The first inequality then simplifies to $4a + 7b \leq 72$. To solve this system, we have to get the signs pointing in the same direction so that we can add the inequalities. Remember that inequalities can be added only if their signs point in the same direction (do not subtract inequalities; think only in terms of adding them). So if we multiply the second inequality by -4, we end up with the following system (note the sign change):

$$\begin{aligned} 4a + 7b &\leq 72 \\ -4a - 4b &\leq -56 \end{aligned}$$

Adding the inequalities, we get $3b \leq 16$, which simplifies to $b \leq 5.33$. Since it's implied that Ava decorates cakes in whole numbers, the maximum possible value of b is 5. This question could've also been solved through guess and check.

15. $\boxed{D}$ For 1 pound of seasoning, Lianne will need 0.75 pounds of sea salt and 0.25 pounds of black pepper. The sea salt will cost $0.75 \times \$2 = \$1.50$ and the black pepper will cost $0.25 \times \$8 = \$2$. Altogether, that's $\$1.50 + \$2 = \$3.50$ for each pound of seasoning. Since Lianne can spend no more than \$210, she is limited to making $\dfrac{\$210}{\$3.50} = 60$ pounds of seasoning. Therefore, 60 is the maximum.

16. $\boxed{54}$ Let s, m, and l represent the numbers of small, medium, and large boxes shipped, respectively. Based on the information given,

$$\begin{aligned} s + m + l &= 250 \\ l &> s + m \end{aligned}$$

Since $m = 70$, the system becomes

$$\begin{aligned} s + l &= 180 \\ l &> s + 70 \end{aligned}$$

Isolating l in the equation gives $l = 180 - s$. Substituting this into the inequality, we get

$$\begin{aligned} 180 - s &> s + 70 \\ 110 &> 2s \\ 55 &> s \end{aligned}$$

Since s is implied to be a whole number, the greatest possible value of s based on this result is 54.

17. $\boxed{7}$ Based on the given information, $n = 10$ and $w = 8x$ (8 ounces of water in each of the x cups). To use these values, we first set up the following inequality:

$$\begin{aligned} C &\leq 16\% \\ \frac{100n}{n + w} &\leq 16 \\ \frac{100(10)}{10 + 8x} &\leq 16 \\ 1000 &\leq 16(10 + 8x) \\ 1000 &\leq 160 + 128x \\ 840 &\leq 128x \\ 6.56 &\leq x \end{aligned}$$

Since the question indicates that x is a whole number, the minimum possible value of x based on this result is 7. It's important to note that we were able to multiply both sides by $10 + 8x$ without worrying about a sign change because $10 + 8x$ is guaranteed to be positive (x must be positive in the context of the problem).